Code No: 111AH **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year Examinations, September - 2023
ENGINEERING DRAWING
(Common to CSE, MIE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) A circle of 50 mm diameter rolls along a straight line without slipping. Draw the curve traced out by a point P on the circumference, for one complete revolution of the circle. Name the curve. Draw a tangent to the curve at a point on it 40 mm from the line.

b) A rectangular plot of land of area 16 sq. m is represented on a map by a similar rectangle of 1 square centimeter. Calculate the RF of the scale of the map. Construct a plain scale to measure up to 60 m and mark a distance of 46 m on the scale. [8+7]

OR

2.a) Draw the curve traced by a point on the circumference of a rolling circle, on a straight line for one and half revolution.

b) On the plan of a shopping complex, a line 10cm long represents a distance of 5m. Draw a diagonal scale for the plan to read up to 6m showing meters, decimetres and centimetres. Mark the lengths 3.24m and 5.57m. [5+10]

3.a) A point A is situated in the first quadrant. Its shortest distance from the intersection point of HP and VP and auxiliary plane is 60 mm and it is equidistant from the principal planes. Draw the projections of the point and determine its distance from the principal planes.

b) The top view of a 75mm long line measures 55 mm. The line is in the VP, its one end being 25 mm above the HP. Draw its projections. [8+7]

OR

4. The top view of a plane is a regular pentagon of side 30 mm having one side inclined at 30° to the VP. Its front view is a straight line inclined at 45° to the reference line. Draw the projections of the plane and determine its true shape. [15]

5. A pentagonal prism is resting on one of the corners of its base in the HP. The longer edge containing that corner is inclined at 30° to HP and the vertical plane containing that edge is inclined at 45° to the VP. Draw the projections of the solid. [15]

OR

6. A hexagonal pyramid of side of base 40 mm and height of axis 110 mm is resting on one of its inclined vertical faces on H.P. such that its axis remains parallel to the V.P. It is cut by a cutting plane which is inclined at an angle 45° with H.P. and bisecting the axis of the pyramid. Draw front view, sectional top view. [15]

7. A pentagonal pyramid of base 30 mm side and height 70 mm, stands with its base on H.P such that one of the base edges is parallel to V.P. It is cut by a section plane perpendicular to V.P, inclined at 30° to H.P and bisecting its axis. Draw the development of the surface of the pyramid. [15]

OR

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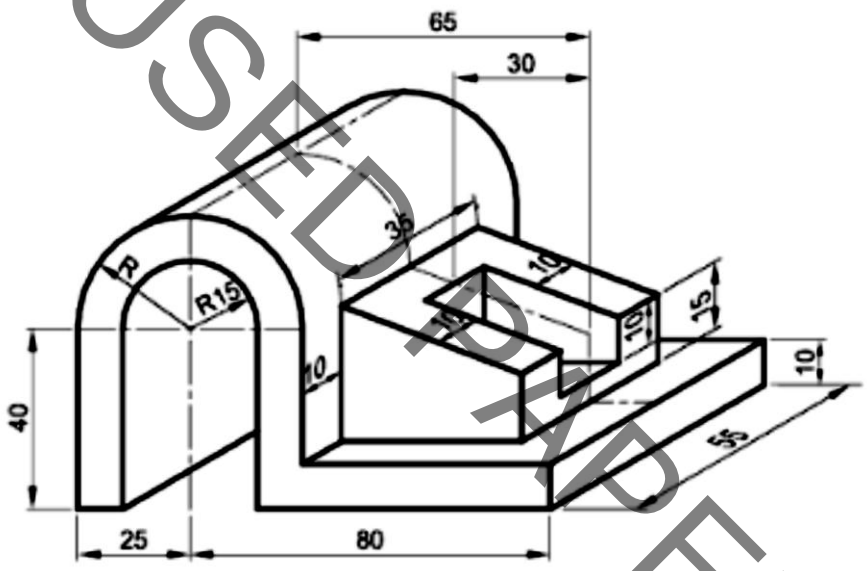
8. A cylinder of base diameter 70 mm is resting on its base on the HP. It is penetrated by another cylinder of base diameter 60 mm, such that their axes intersect each other at right angles. Draw the projections of the combination and show the curves of intersections. [15]

9.a) A vertical cylinder of base diameter 50 mm and height 70 mm is cut by a plane inclined at 55° to HP and perpendicular to VP, which meets the axis at a distance of 20 mm from top base. Draw the isometric view of the remaining portion of the cylinder.

b) A sphere of 50 mm diameter is surrounded centrally on the top of a square block of side 60 mm and thickness 20 mm. Draw the isometric view of the arrangement. [8+7]

OR

10. Draw the front view, side view and top view for the given figure. [15]



All dimensions are in mm

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